WORKED SOLUTION

STEP I 2018, Question 1

Pure • Area Between Curves

STEP I 2018, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q1

The key idea

Setting the cubic equal to the line gives a factor of x and then a difference of two squares, so all three intersections come out cleanly. After that everything is routine calculus, and the substitution $c = b - a$ (the x-coordinate of P) keeps the algebra tidy. The final inequality looks daunting until both areas are written in terms of c: then it collapses to $b > a$.

Solution

The intersections and the sketch

At an intersection, $x(b-x)^2 = a^2x$, so either $x = 0$ (the origin) or

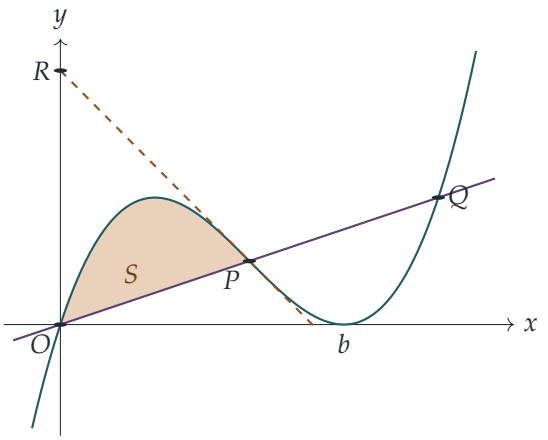
$$(b-x)^2 = a^2 \quad \Longrightarrow \quad b - x = \pm a \quad \Longrightarrow \quad x = b - a \ \text{ or } \ x = b + a.$$

Since $a > 0$, $b - a < b + a$, so

$$P = (b-a,\ a^2(b-a)), \qquad Q = (b+a,\ a^2(b+a)).$$

Because $0 < a < b$, both points lie in the first quadrant.

For the sketch: the cubic $y = x(b-x)^2$ has a single root at O and a repeated root at $x = b$, where it touches the x-axis. It is negative for $x < 0$, positive for $x > 0$ (apart from $x = b$), and tends to $+\infty$ as $x \to \infty$. Its derivative (found below) is $(b-x)(b-3x)$, so it has a local maximum at $x = b/3$ and a local minimum at $(b, 0)$. The line through O cuts the rising branch near O, crosses again at P (between $b/3$ and b when $a < \frac{2}{3}b$, or before $b/3$ otherwise), and meets the curve a third time at Q, beyond the touching point $x = b$. The figure uses $a = 1$, $b = 3$; it also shows the tangent at P, the point R, and the region of area S.



The tangent at P

Differentiating with the product rule,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = (b-x)^2 - 2x(b-x) = (b-x)(b-3x).$$

At P, $x = b-a$, so $b-x = a$ and $b-3x = 3a-2b$. The gradient there is $m = a(3a-2b)$.

The tangent is $y = mx + k$, where k is chosen so that the line passes through P:

$$\begin{aligned} k &= a^2(b-a) - a(3a-2b)(b-a) \\ &= a(b-a)(a-3a+2b) \\ &= 2a(b-a)^2. \end{aligned}$$

Hence the tangent at P is $y = a(3a-2b)x + 2a(b-a)^2$, as required. In particular $R = (0,\, 2a(b-a)^2)$, which lies on the positive y-axis.

The area S

Write $c = b-a > 0$ for the x-coordinate of P. On $0 < x < c$,

$$x(b-x)^2 - a^2x = x\,(b-x-a)(b-x+a) > 0,$$

since all three factors are positive there. So the curve lies above the line between O and P, and

$$S = \int_0^c (x(b-x)^2 - a^2x)\,\mathrm{d}x = \int_0^c ((b^2-a^2)x - 2bx^2 + x^3)\,\mathrm{d}x = \frac{(b^2-a^2)c^2}{2} - \frac{2bc^3}{3} + \frac{c^4}{4}.$$

Now substitute $b^2 - a^2 = c(b+a) = c(c+2a)$ and $b = c+a$:

$$\begin{aligned} S &= \frac{c^3(c+2a)}{2} - \frac{2c^3(c+a)}{3} + \frac{c^4}{4} \\ &= c^3\Big[\,c(\tfrac{1}{2} - \tfrac{2}{3} + \tfrac{1}{4}) + a(1-\tfrac{2}{3})\Big] \\ &= c^3\Big(\frac{c}{12} + \frac{a}{3}\Big) = \frac{c^3(c+4a)}{12}. \end{aligned}$$

Since $c + 4a = b + 3a$, this is

$$S = \frac{1}{12}(b-a)^3(3a+b),$$

as required.

Ask yourself Why not just expand everything in a and b?

You can, but the target has a factor $(b-a)^3$, and pulling it out of a quartic in a and b is painful. Writing $c = b - a$ makes that factor appear automatically, because every term of the integral already carries c^2 or more.

The triangle T and the inequality

Take OR as the base of triangle OPR: it lies along the y-axis with length $2ac^2$. The perpendicular height is the distance from P to the y-axis, namely c. So

$$T = \tfrac{1}{2} \cdot 2ac^2 \cdot c = ac^3 = a(b-a)^3.$$

Therefore, since $(b-a)^3 > 0$,

$$S - \tfrac{1}{3}T = \frac{(b-a)^3}{12}(3a + b - 4a) = \frac{(b-a)^4}{12} > 0,$$

because $b \neq a$. Hence $S > \frac{1}{3}T$, as required.

Checks

- With $a = 1$, $b = 3$ (the figure): $P = (2,2)$, $Q = (4,4)$, and the curve gives $y = 2$ at $x = 2$ and $y = 4$ at $x = 4$. The tangent formula gives $y = -3x + 8$; the gradient formula $(3-x)(3-3x)$ at $x = 2$ is -3. Integrating directly, $S = \int_0^2 (x^3 - 6x^2 + 8x)\,\mathrm{d}x = 4 - 16 + 16 = 4$, matching $\frac{1}{12} \cdot 8 \cdot 6 = 4$. Also $T = 8$, and $4 > \frac{8}{3}$.
- The ratio $S/(\frac{1}{3}T) = (3a+b)/(4a)$ tends to 1 as $b \to a$, so the constant $\frac{1}{3}$ is the best possible. That makes sense: for a parabola $y = px - qx^2$ the same construction gives $S = qc^3/6$ and $T = qc^3/2$, so $S = \frac{1}{3}T$ exactly, and when P is very close to O the cubic between them is almost a parabola.
- For thousands of random pairs $0 < a < b$, a computer check of the intersections, the tangent, the area (by numerical integration) and the inequality agreed with the formulas.

Common slips

- Dividing $x(b-x)^2 = a^2 x$ by x without noting the root $x = 0$, or taking only $b - x = a$ and losing Q.

- Integrating line minus curve and getting a negative "area": check which graph is on top before integrating.
- In the sketch, drawing the cubic crossing the x-axis at $x = b$ rather than touching it there.

Beyond STEP

The limiting ratio of 1 in the last part is in the spirit of Archimedes' quadrature of the parabola: for a parabola the segment has exactly a third of the area of triangle OPR, and near O the cubic is locally a parabola, so the inequality sharpens to equality in the limit. Understanding *why* local behaviour is governed by the first few Taylor terms, and making "almost a parabola" precise, is the business of first-year analysis. Abbott's *Understanding Analysis* is a friendly place to see Taylor's theorem done properly.

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